

CS451: Spring 2026

10 Feb 2026

(This is an ungraded test)

Question 1. Difficulties (3 marks)

Give 3 difficulties associated with parallel or distributed systems:

1.
2.
3.

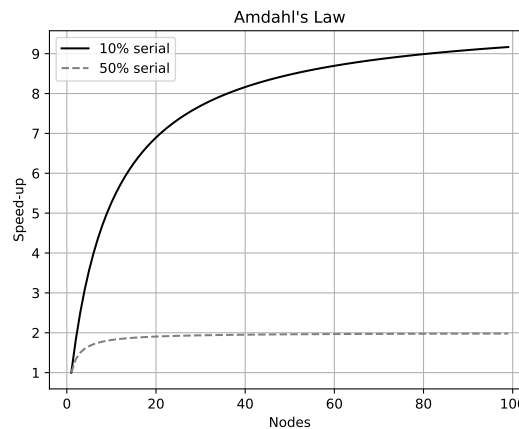
Question 2. Reliability (3 marks)

Distributed systems can be *more* reliable than non-distributed systems—why?
In your answer, give an example of a specific distributed system.

Answer.

Question 3. Amdahl's Law (3 marks)

Consider this graphing of Amdahl's model for $\sigma = 0.1$ and $\sigma = 0.5$.



What is the intuition behind this model?

Answer.

Question 4. Multi-threading (9 marks)

Consider the following code:

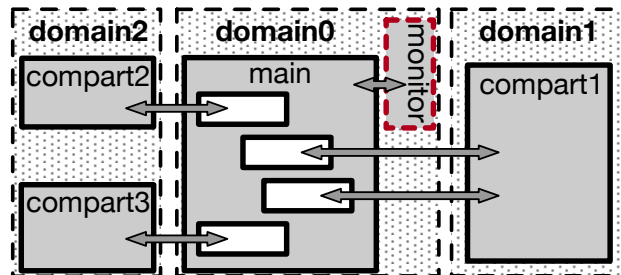
```
1 // From CS:APPe3
2 #include <pthread.h>
3 #include <stdio.h>
4 #include <stdlib.h>
5
6 volatile long cnt = 0; /* Counter */
7
8 void *thread(void *vargp)
9 {
10     long i, niters =
11         *((long *)vargp);
12     for (i = 0; i < niters; i++)
13         cnt++;
14     return NULL;
15 }
16
17 int main(int argc, char **argv)
18 {
19     long niters;
20     pthread_t tid1, tid2;
21     niters = atoi(argv[1]);
22     pthread_create(&tid1, NULL, thread, &niters);
23     pthread_create(&tid2, NULL, thread, &niters);
24     pthread_join(tid1, NULL);
25     pthread_join(tid2, NULL);
26     /* Check result */
27     if (cnt != (2 * niters))
28         printf("BOOM! cnt=%ld\n", cnt);
29     else
30         printf("OK cnt=%ld\n", cnt);
31     exit(0);
32 }
```

The table below references definitions in the program shown above. Tick ☒ the box that indicates whether a specified definition is *private* to a thread, or *shared* among more than one thread:

Definition	Is Private?	Is Shared?
cnt (line 6)	☐	☐
vargp (line 8)	☐	☐
i (line 10)	☐	☐
niters (line 10)	☐	☐
niters (line 19)	☐	☐
argc (line 17)	☐	☐
argv (line 17)	☐	☐
tid1 (line 20)	☐	☐
tid2 (line 20)	☐	☐

Question 5. Compartmentalized Software (4 marks)

Consider this abstract diagram showing a distributed system consisting of compartmentalized software:



1. What is the benefit of compartmentalizing software?

Answer.

2. What does “IPC” stand for?

Answer:

3. What kind of IPC would be used between compartments in the diagram shown above?

☐ Message-passing

☐ Shared memory

Rough Notes / Scrap Page

Do not detach this (or any) page from the exam booklet.